

Lab 04 — Probabilistic Models, Built in JAX

Fundamentos de Aprendizagem Automática

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Overview

Class 04 · Topic 3 (second half). Lecture: `slides/04_probabilistic_models.pdf`. Demo notebooks in `notebooks/04-probabilistic-models/`: `01_mle_map.ipynb`, `02_naive_bayes_classification.ipynb` and `03_naive_bayes_regression.ipynb`.

Notebook for this lab: `notebooks/04-probabilistic-models/lab04_probabilistic_` with helpers in `lab04_utils.py` (same folder, do not edit). Cells marked **TODO** contain `raise NotImplementedError`: replace it with your code. Every other cell is ready, and it draws the plots and prints the comparisons once your functions work.

Solution: `solutions/lab04_probabilistic_models_solution.ipynb` is the fully solved notebook. Before each function it explains how the solution was reached, and after each experiment it answers this guide's questions with the numbers the run produces. Try every task yourself first.

The rule of this lab. Where a model needs an optimizer, it is *a negative log-likelihood written with `jax.numpy` and the gradient comes from `jax.grad` (through `train`)*. Naive Bayes needs no gradient: its training is **counting**, and JAX scores all examples at once, in log space. Every model is compared with its scikit-learn twin and the notebook prints the largest difference.

By the end of the lab you should be able to:

1. Derive a loss as a negative log-likelihood, add a prior, and recognize ridge and Laplace smoothing as MAP estimates.
2. Build Naive Bayes for **discrete** (Bernoulli, categorical) and **continuous** (Gaussian) features by counting, with smoothing.
3. Use Naive Bayes for **regression**, with a binned target and with a continuous

target, and explain where the independence assumption hurts.

4. Measure the quality of a probability (log-loss, Brier score, reliability), calibrate it, and make cost-based decisions.

Part	Task	twin	Time
A	MLE and MAP with <code>jax.grad</code> : Gaussian, Beta prior, ridge as MAP	Ridge, BernoulliNB (alpha)	20 min
B	Naive Bayes, discrete features: Bernoulli, Laplace smoothing	BernoulliNB	15 min
C	Naive Bayes, continuous features: Gaussian; categorical (binned)	GaussianNB, CategoricalNB	20 min
D	Naive Bayes for regression : binned target; continuous target	GaussianNB, CategoricalNB, LinearRegression	35 min
E	Probabilities: log-loss, Brier, reliability; decisions with costs	<code>log_loss</code> , <code>brier_score_loss</code> , <code>calibration_curve</code>	20 min
F	Challenge: Platt scaling with <code>jax.grad</code>	LogisticRegression	home

Setup

```
cd faa
```

```
source venv/bin/activate
```

```
# see README: make venv, or pip install .
jupyter lab notebooks/04-probabilistic-models/lab04_probabilistic_models.
```

The data are those of Class 03: **spam** (Spambase, `load_spam()`) for classification and **house prices** (California housing, `load_housing()`) for regression. The notebook prepares four views of the spam features: `B_s` “the word appears” (binary), `L_s` $\log(1 + x)$ (continuous), `Z_s` four quantile bins of `L_s` (categorical), and the raw `X_s`. Scaling statistics and bin edges come from the **training** part only. `stratified_split`, `fit_standardizer`, `standardize`, `classification_metrics` and `regression_metrics` are your Class 03 functions, given ready-made.

Part A — MLE and MAP with `jax.grad`

A1. Gaussian maximum likelihood

Implement `gaussian_nll(params, x)`, the negative log-likelihood of $x_1, \dots, x_n \sim \mathcal{N}(\mu, \sigma^2)$, with `params = {"mu": ..., "log_sigma": ...}`:

$$-\ell(\mu, \sigma) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2$$

The driver trains it on five numbers and compares with the sample mean and with `np.var`, then measures the bias of the MLE variance by simulation.

1. Which variance does gradient descent find: the divide-by- n or the divide-by- $(n - 1)$ one? Why?
2. What is the average MLE variance over many samples of size 5 from $\mathcal{N}(0, 1)$, and why is it not 1?

A2. MAP for a coin: the Beta prior

Implement `log_posterior_bernoulli(theta, k, n, a, b) = (k + a - 1) \log \theta + (n - k + b - 1) \log(1 - \theta)` for a scalar θ . The driver maps it over a grid with `jax.vmap` and compares the maximum with the closed form $(k + a - 1)/(n + a + b - 2)$, then compares `BernoulliNB(alpha)` with a MAP estimate.

1. Compare MLE, Beta(2,2) MAP and Beta(10,10) MAP after 3 heads in 3 tosses, and after 30 tosses with 21 heads. What does the prior do, and when does it stop mattering?
2. Which prior does `BernoulliNB(alpha)` correspond to?

A3. MAP for regression: ridge is a Gaussian prior

Implement `map_loss(params, X, y, sigma2, tau2)`, the negative log-posterior without constants, $\frac{1}{2\sigma^2} \sum_i (y_i - w^\top x_i - b)^2 + \frac{1}{2\tau^2} \|w\|^2$ (no prior on the bias), and compare its minimizer with `Ridge(alpha = sigma2 / tau2)`.

1. How close are the two? What does σ^2/τ^2 mean as a regularization strength?

Part B — Naive Bayes with Discrete Features

$$\hat{c} = \arg \max_c \left[\log P(c) + \sum_j \log P(x_j | c) \right]$$

B1. Bernoulli Naive Bayes

Binary features $b_j = [x_j > 0]$. Implement `fit_bernoulli_nb(Xb, y, alpha, n_classes=2)` returning the log-prior of each class and the logs of $\theta_{cj} = (n_{cj} + \alpha)/(n_c + 2\alpha)$ and of $1 - \theta_{cj}$; `joint_log_bernoulli(model, Xb)` returning $\log P(c) + \log P(x | c)$ for every example and class, shape (n, K) (use `jnp.where(Xb == 1, log_theta, log_one_minus)`, broadcast and summed: it avoids $0 \cdot (-\infty)$); and `posterior_proba(joint)`, the normalized $P(c | x)$ computed with `logsumexp`.

The driver compares the joint log-probabilities and the posteriors with `BernoulliNB(alpha=1)`, prints the metrics, and counts the test e-mails that get an *impossible* class $(-\infty)$ with $\alpha = 0$ and with $\alpha = 1$.

1. How close are you to scikit-learn, and how good is the classifier?
2. What does $\alpha = 0$ do with 20 training e-mails, and with 100? Why does smoothing matter most for rare words?

Part C — Naive Bayes with Continuous Features

C1. Gaussian Naive Bayes

Implement `fit_gaussian_nb(X, y, var_smoothing=1e-9, n_classes=2)` returning the log-prior, the class means and the class variances (the maximum-likelihood variance, **dividing by n** ; add `var_smoothing * max_j Var(x_j)`

to every variance), and `joint_log_gaussian(model, X)`. Broadcast the examples $(n, 1, d)$ against the class parameters $(1, K, d)$: no loops over examples. Use the log features `L_s`.

1. How close are you to GaussianNB? Compare its metrics with those of Bernoulli NB and explain its high recall and low precision.

C2. Categorical Naive Bayes

Implement `fit_categorical_nb(Z, y, alpha, n_categories, n_classes=2)` returning the log-prior and $\log \theta_{cjm} = \log \frac{n_{cjm} + \alpha}{n_{cj} + \alpha M}$ (shape (K, d, M) , use `np.bincount`), and `joint_log_categorical(model, Z)`. The driver compares with `CategoricalNB(min_categories=4)` and then trains logistic regression and the three Naive Bayes models on the first n training e-mails, for growing n , and plots the test accuracy.

1. Which Naive Bayes model is best on spam? Why is the categorical one not clearly better than the Bernoulli one?
2. Which model wins with 20 examples, and which with 3680? Why do the NB curves plateau?

Part D — Naive Bayes for Regression

The target is now a number. Two routes.

D1. Discretize the target

Implement `bin_target(y, n_bins)` returning the interior quantile **edges** of y (`np.quantile`) and the **bin means** m_k (the class of a value is `np.searchsorted(edges, y, side="right")`), and `predict_expected(joint, means)`, the expected value $\hat{y} = \sum_k P(k | x) m_k$ from the joint log-scores of a classifier (use `posterior_proba`).

The driver trains your Gaussian NB on the bin labels for $K = 5, 10, 20$, predicts the expected value and the most probable bin, and compares with `KBinsDiscretizer + GaussianNB`.

1. How do your bin labels and expected values compare with scikit-learn's?
2. Expected value or most probable bin? Compare their R^2 with linear regression's and explain.

D2. Discrete features for regression

Discretize the **features** as well (`fit_bins / apply_bins`, 3, 5, 10 or 20 quantile bins) and use your categorical NB with the K target bins.

1. How does R^2 change with the number of bins per feature? Compare with the Gaussian version and with linear regression.

D3. A continuous target: linear-Gaussian Naive Bayes

$$y \sim \mathcal{N}(\mu_y, \tau^2), \quad x_j \mid y \sim \mathcal{N}(a_j + b_j y, s_j^2) \text{ (independent given } y)$$

Implement `fit_lgnb(X, y)` returning $(\mu_y, \tau^2, a, b, s^2)$: the mean and variance of y , the least-squares fit of every feature **on** y (`np.linalg.lstsq` on $[y, \mathbf{1}]$) and the residual variances. Then `predict_lgnb(model, X)` returning the posterior mean and standard deviation of y :

$$P = \frac{1}{\tau^2} + \sum_j \frac{b_j^2}{s_j^2}, \quad \hat{y} = \frac{1}{P} \left(\frac{\mu_y}{\tau^2} + \sum_j \frac{b_j (x_j - a_j)}{s_j^2} \right), \quad \text{sd} = \frac{1}{\sqrt{P}}$$

The driver **verifies your formula without it**: the negative log-posterior is written in `jax.numpy`, `jax.grad` climbs it for five districts, and the maximum must equal your posterior mean. It then prints R^2 and the coverage of the 90% interval.

1. Does the closed form agree with the gradient ascent?
2. How good is the model against linear regression and against D1? What is the posterior standard deviation, and does it depend on the district?

D4. Where the naive assumption hurts

The driver appends 0, 1, 3, 10 exact copies of the most informative feature (median income) and refits linear regression and your linear-Gaussian NB.

1. What happens to each model's R^2 , to the posterior standard deviation and to the coverage of the 90% interval? Explain in terms of the independence assumption.

Part E — Probabilities and Decisions

The driver trains a logistic regression on the standardized log features to compare with your Naive Bayes models.

E1. Log-loss, Brier score and reliability

Implement `log_loss(y, p) =  $-\frac{1}{n} \sum [y \log p + (1 - y) \log(1 - p)]$` (clip p to $[10^{-15}, 1 - 10^{-15}]$), `brier_score(y, p) =  $\frac{1}{n} \sum (p - y)^2$` and `reliability_table(y, p, bins)`, which cuts $[0, 1]$ into equal bins and returns, for every **non-empty** bin, the mean predicted probability, the observed frequency of spam and the count. The driver checks them against `log_loss`, `brier_score_loss` and `calibration_curve`.

1. Compare the three models on accuracy, log-loss and Brier score. What does the log-loss of Gaussian NB say that its accuracy does not?
2. Read the reliability diagrams.

E2. Decisions with costs

A false positive (a real e-mail hidden) costs c_{FP} , a false negative (a spam delivered) costs c_{FN} . Implement `bayes_threshold(c_fp, c_fn) =  $c_{FP} / (c_{FP} + c_{FN})$` and `expected_cost(y, p, threshold, c_fp, c_fn)`, the mean cost per e-mail. The driver compares, for the costs 1:1, 10:1 and 1:5, the threshold 0.5, the Bayes threshold and the best threshold found by scanning the test set.

1. How much does the cost fall when the threshold comes from the costs? Is the theoretical threshold the best one?
2. What happens to the cost at the Bayes threshold if the probabilities come from Gaussian NB?

Part F — Challenge: Platt Scaling with jax.grad

Repair the probabilities of the Gaussian NB. **Platt scaling** learns a logistic map $p = \sigma(at + b)$ from a transform t of the classifier's score $s = \log P(\text{spam} \mid x) - \log P(\text{ham} \mid x)$.

1. Split the training set: the first 75% fits the Gaussian NB, the last 25% is the **calibration set**.

2. The Naive Bayes log-odds reach $\pm 10^8$: compress them with $t = \text{sign}(s) \log(1 + |s|)$ (monotone). Implement `platt_loss(params, t, y)` (the binary cross-entropy of $\sigma(at + b)$, `params = {"a", "b"}`) and fit it on the calibration set with `train`.
3. On the test set report the log-loss, the Brier score, the accuracy and the cost at the Bayes threshold for 10:1 costs, before and after calibration. Compare your (a, b) with a `LogisticRegression` fitted on the same one-dimensional score.

Why must the calibration data differ from the training data? What did calibration repair, and what did it not?

Going further. Replace Platt scaling by isotonic regression (`sklearn.isotonic.IsotonicRegression` or use `CalibratedClassifierCV`, and compare. Or build a **multinomial** Naive Bayes for word *counts* (the natural model for text) and compare it with the Bernoulli one.

Checklist

- ☐ Your Gaussian MLE, the Beta-prior MAP and the ridge MAP match the closed forms and `scikit-learn`.
- ☐ Your Naive Bayes models (Bernoulli, Gaussian, categorical) match `scikit-learn`'s joint log-probabilities, and you can explain Laplace smoothing as a prior.
- ☐ You can turn a classifier into a regressor (binned target, expected value), and your linear-Gaussian posterior equals the maximum found by `jax.grad`.
- ☐ You can explain why duplicating a feature makes Naive Bayes overconfident.
- ☐ You can compute log-loss, Brier score and a reliability table, and you know why accuracy is not enough for probabilities.
- ☐ You can derive the cost-based threshold and know that it needs calibrated probabilities.